Algorithm for frequency analysis



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1) Input data:

1.1 Global system matrices:

- Stiffness matrix: $[\mathbf{K}]$
- Mass matrix: $[\mathbf{M}]$

Tip: For increased performance in Matlab, initialize $[\mathbf{K}]$ and $[\mathbf{M}]$ as **sparse** matrices before the assembly (i.e., use the **sparse** function instead of the **zeros** function):

```
K = sparse(Ndof,Ndof);
M = sparse(Ndof,Ndof);
```

1.2 Global forces vector:

- External forces amplitudes:

$$[\mathbf{F}] = [\cdots \quad \widehat{\boldsymbol{F}}_k \quad \cdots]$$ (as many rows as DOFs of the problem)

- Frequencies:

$$\{\boldsymbol{\omega}\} = \{\cdots \quad \omega_k \quad \cdots\}$$

1.3 Boundary conditions:

- Fix nodes and DOFs matrix:

$$[\mathbf{U}_\mathrm{p}] = \begin{bmatrix} \vdots & & \\ u^{(p)} & n^{(p)} & j^{(p)} \\ \vdots & & \end{bmatrix}$$ } Number of prescribed DOFs

$u^{(p)}$: value of prescribed displ./rot. (p)
$n^{(p)}$: global node # assigned to (p)
$j^{(p)}$: degree of freedom assigned to (p)

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2) Boundary conditions:

2.1 Initialization:

$[\mathbf{U}] = [\mathbf{0}]_{N_{\text{dof}} \times N_\omega}$ (N_ω is the total number of frequencies)

2.2 Prescribed and free DOFs:

For each row p in $[\mathbf{U}_\text{p}]$

$\boldsymbol{I}_\text{p}(p) = 6\big(\mathbf{U}_\text{p}(p,2) - 1\big) + \mathbf{U}_\text{p}(p,3)$ (vector with prescribed degrees of freedom)

$\mathbf{U}\big(\boldsymbol{I}_\text{p}(p), :\big) = \mathbf{U}_\text{p}(p,1)$

End loop over rows in $[\mathbf{U}_\text{p}]$

$\boldsymbol{I}_\text{f} = \{1 : N_\text{dof}\} - \{\boldsymbol{I}_\text{p}\}$ (**Tip**: in Matlab, this operation can be done with the **`setdiff`** function)

```
If = setdiff(1:Ndof,Ip)
```

3) Solve system of equations:

3.1 Solve system:

For each column k in $[\mathbf{F}]$

$\mathbf{U}(\boldsymbol{I}_\text{f}, k) = [\mathbf{K}(\boldsymbol{I}_\text{f}, \boldsymbol{I}_\text{f}) - \omega(k)^2 \mathbf{M}(\boldsymbol{I}_\text{f}, \boldsymbol{I}_\text{f})]^{-1} \big(\mathbf{F}(\boldsymbol{I}_\text{f}, k) - \big[\mathbf{K}\big(\boldsymbol{I}_\text{f}, \boldsymbol{I}_\text{p}\big) - \omega(k)^2 \mathbf{M}\big(\boldsymbol{I}_\text{f}, \boldsymbol{I}_\text{p}\big)\big]\{\mathbf{U}\big(\boldsymbol{I}_\text{p}, k\big)\}\big)$

End loop over columns in $[\mathbf{F}]$

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4) Modal analysis:

4.1 Solve eigenvalues and eigenvectors problem. In Matlab, for **`sparse`** stiffness and mass matrices, use the following command to obtain the first **`Nm`** eigenvectors (**V**) associated to the smallest eigenvalues (**D**) in magnitude:

```
[V,D] = eigs(K(If,If),M(If,If),Nm,'sm');
```

Note: For non-sparse (full) matrices, use `eig` function instead:

```
[V,D] = eig(K(If,If),M(If,If));
```

Tip: Make sure $[\mathbf{K}]$ and $[\mathbf{M}]$ are symmetric before using `eigs`/`eig` functions:

$[\mathbf{K}] = \left([\mathbf{K}] + [\mathbf{K}]^{\mathrm{T}}\right)/2$

$[\mathbf{M}] = \left([\mathbf{M}] + [\mathbf{M}]^{\mathrm{T}}\right)/2$

4.2 Obtain squared natural frequencies (λ_k) and vibration modes ($\widehat{\boldsymbol{\phi}}_k$):

$[\boldsymbol{\Phi}] = [\mathbf{0}]_{N_{\mathrm{dof}} \times N_{\mathrm{m}}}$

$\{\boldsymbol{\lambda}\} = \{\mathbf{0}\}_{1 \times N_{\mathrm{m}}}$

For each column k in $[\mathbf{V}]$

$\boldsymbol{\Phi}(\boldsymbol{I}_{\mathrm{f}}, k) = \mathbf{V}(:,k)/\mathrm{sqrt}\left(\{\mathbf{V}(:,k)\}^{\mathrm{T}}[\mathbf{M}(\boldsymbol{I}_{\mathrm{f}}, \boldsymbol{I}_{\mathrm{f}})]\{\mathbf{V}(:,k)\}\right)$ (mass-normalized vibration mode)

$\lambda(k) = \mathbf{D}(k,k)$ ($\equiv \omega^2$ squared natural frequency)

End loop over columns in $[\mathbf{V}]$

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5) Model-order reduction:

5.1 Select set of modes to project the system:

$\{\boldsymbol{I}_\mathrm{m}\} = \{\cdots\}$ ($\boldsymbol{I}_\mathrm{m}$ can be any set of numbers between $[1, N_\mathrm{m}]$, including all of them)

5.2 Project the system onto the selected set of modes:

$[\mathbf{U}^*] = [\mathbf{0}]_{N_\mathrm{dof} \times N_\omega}$ (approximate solution vector)

For each column k in $[\mathbf{F}]$

For each mode j in $\{\boldsymbol{I}_\mathrm{m}\}$

$\boldsymbol{\alpha}(j,k) = \left\{\boldsymbol{\Phi}\left(:, \boldsymbol{I}_\mathrm{m}(j)\right)\right\}^\mathrm{T} \{\mathbf{F}(:,k)\} / (\lambda(j) - \omega(k)^2)$ (modal amplitude)

$\mathbf{U}^*(:,k) = \mathbf{U}^*(:,k) + \boldsymbol{\Phi}\left(:, \boldsymbol{I}_\mathrm{m}(j)\right)\boldsymbol{\alpha}(j,k)$

End loop over modes

End loop over columns in $[\mathbf{F}]$

Note: The model-order reduction procedure is also applicable to static problems, just by setting $\omega(k) = 0$ and making $\{\mathbf{F}(:,k)\}$ the corresponding (static) force vector.